ASSIGNMENT 4

Unit Covered	Unit IV — Motion Representation
Suggested Tools	Python (OpenCV — Lucas-Kanade & Farneback), a short video clip

1. Record or source a short video clip (5–10 seconds) with clear motion — a rolling/thrown ball, a walking person, or moving traffic all work.
2. Extract two consecutive frames and explain, using the brightness constancy equation, the core assumption optical flow relies on — and describe one real situation in your clip where this assumption might break down (e.g., reflections, motion blur, lighting change).
3. Implement sparse optical flow using the Lucas-Kanade (feature-based) method on a handful of tracked points, and dense optical flow using the Farneback (differential) method on the full frame. Visualise both as vector-field overlays on the frame.
4. Compare the two outputs: where does the sparse method fail to give useful vectors, and where does the dense method give noisy/unreliable vectors? Support your answer with cropped image evidence.
5. Estimate the affine flow parameters for a small rigid patch in your video (e.g., a car body or a person's torso) and interpret what each recovered parameter means physically (translation, rotation, scaling components).
6. Add regularization (e.g., a smoothness constraint) to your dense flow estimate conceptually or practically, and discuss how it changes the result in textureless regions of the frame.